Physics 3221 Spring 2025: Midterm 2

Grade:

1. Mystery Force and its Potential (20 Points)

Consider the following force: $\vec{F} = ky\hat{x} + kx\hat{y}$.

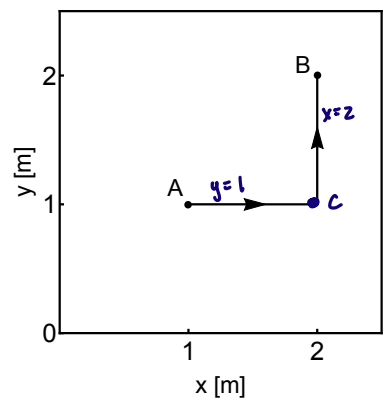


Figure 1: The path the particle follows going from point A to point B

(a) Calculate $\vec{\nabla} \times \vec{F}$. Is this force conservative?

$$(\vec{\nabla}\times\vec{F})_x = \frac{\partial F_z}{\partial y} - \frac{\partial F_y}{\partial z} = 0 - 0 = 0$$

$$(\vec{\nabla}\times\vec{F})_y = \frac{\partial F_x}{\partial z} - \frac{\partial F_z}{\partial x} = 0 - 0 = 0$$

$$(\vec{\nabla}\times\vec{F})_z = \frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} = \frac{\partial}{\partial x}(kx) - \frac{\partial}{\partial y}(ky) = k - k = 0$$

$$\boxed{\vec{\nabla}\times\vec{F} = 0 \Rightarrow \text{Conservative}}$$

(b) Find the W the force does on a particle moving from point A to point B along the path depicted in Fig. 1.

$$W = \int_A^B \vec{F}\cdot d\vec{r} = \int_A^B (F_x dx + F_y dy) = \int_A^C (F_x dx + \cancel{F_y dy}^{\,0}) + \int_C^B (\cancel{F_x dx}^{\,0} + F_y dy)$$

$d\vec{r} = dx\,\hat{x}$ $\qquad$ $d\vec{r} = dy\,\hat{y}$

$$= k\int_{1m}^{2m} y\,dx + k\int_{1m}^{2m} x\,dy = k\int_{1m}^{2m} (1)\,dx + k\int_{1m}^{2m} (2)\,dy$$

$$= kx\Big|_{1m}^{2m} + k2y\Big|_{1m}^{2m} = [k(2-1) + 2(2-1)k]\,m$$

$$= [k + 2(1)k]\,m = \boxed{3km = W}$$

(c) For this force, $U(x, y) = -kxy$. Verify that $\vec{F} = -\vec{\nabla}U$

$$\vec{\nabla}U = \frac{\partial}{\partial x}(-kxy)\,\hat{x} + \frac{\partial}{\partial y}(-kxy)\,\hat{y}$$

$$= -ky\,\hat{x} - kx\,\hat{y}$$

$$= -(ky\,\hat{x} + kx\,\hat{y}) = -\vec{F}$$

$$\Rightarrow \boxed{\vec{F} = -\vec{\nabla}U}$$

Grade:

2. The Underwater Clock

A scuba diver finds a submerged clock sitting inside a sunken ship. The clock tells time using a swinging pendulum, see Fig. 2. Assume the mass attached to the pendulum experiences a gravitational force with force constant g, and ignore the buoyancy force.

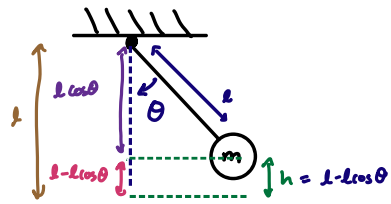


Figure 2: The pendulum inside the underwater clock. The length of the pendulum arm is l and the mass attached is m.

(a) Neglecting drag, show that the potential energy of the mass on the pendulum for small angles $\theta \ll 1$ is $U(\theta) \approx \frac{1}{2}k\theta^2$. Find k in terms of m and l.

$$U = mgh = mgl(1-\cos\theta) \approx mgl\left[1-\left(1-\tfrac{1}{2}\theta^2+\dots\right)\right] \approx mgl\left[1-1+\tfrac{1}{2}\theta^2\right] = \tfrac{1}{2}mgl\theta^2$$
$$= \tfrac{1}{2}k\theta^2, \quad k = mgl$$

(b) Find the net force acting on the mass in the angular direction, again assuming small angels and ignoring terms of $\mathcal{O}(\theta^3)$.

$$\vec{F} = -\vec{\nabla}U \Rightarrow F_\theta = \underbrace{-\frac{1}{r}\frac{\partial}{\partial\theta}}_{\text{polar coord.}}\left(\tfrac{1}{2}mgl\theta^2\right)\Big|_{\underbrace{r=l}_{\text{the pendulum always keeps } r=l \text{ = constant}}} = -\frac{1}{l}(mgl\theta) = -mg\theta$$

(c) Write down the equation of motion for the pendulum, assuming $\theta \ll 1$.

$$F_\theta = mr\ddot{\theta} + 2\dot{r}\dot{\theta} = ml\ddot{\theta} + 0 \quad \left\{\begin{array}{l}\dot{r}=0 \text{ for the pendulum}\\ r=l\end{array}\right.$$

$$\boxed{-mg\theta = ml\ddot{\theta}}$$

Rewrite: $\ddot{\theta} = -g/l\,\theta \Rightarrow \ddot{\theta} = -\omega_0^2\theta, \quad \omega_0 \equiv \sqrt{g/l}$

(d) Find $\theta(t)$. Use the following initial conditions: at $t = 0$, the pendulum starts at an angle θ_0 and is released from rest.

we know $\frac{d^2}{dt^2}(\cos\omega_0 t) = \frac{d}{dt}(-\omega_0\sin\omega_0 t) = -\omega_0^2\cos\omega_0 t$

$\frac{d^2}{dt^2}(\sin\omega_0 t) = \frac{d}{dt}(\omega_0\cos\omega_0 t) = -\omega_0^2\sin\omega_0 t$

$\Rightarrow$ $\theta(t) = B_1\cos\omega_0 t + B_2\sin\omega_0 t$

At $t=0$, $\theta = \theta_0$ $\Rightarrow$ $\theta_0 = B_1$

at $t=0$, $\dot{\theta} = 0$ $\Rightarrow$ $\dot{\theta} = 0 = -B_1\omega_0\sin\omega_0 t + B_2\omega_0\cos\omega_0 t$

$0 = 0 + B_2\omega_0(1) \Rightarrow B_2 = 0$

$\Rightarrow \boxed{\theta(t) = \theta_0\cos\omega_0 t}$

(e) What is the frequency and amplitude of oscillations?

frequency $= \omega_0 = \sqrt{g/l}$

amplitude $= \theta_0$

Grade:

3. The Damped Underwater Clock

Now, consider the effect of drag from the water on the mass in the pendulum shown in Fig. 2. The water exerts a drag force $\vec{F}_{\text{drag}} = -b\vec{v}$.

(a) Show that the new equation of motion for the pendulum is:

$$\ddot{\theta} + 2\beta\dot{\theta} + \omega_0^2\theta = 0$$

$\vec{F}_{drag} = -b\vec{v} = -b(\underset{r=l}{r}\dot{\theta}\hat{\theta} - \underset{=0}{\dot{r}}\hat{r}) = -bl\dot{\theta}\hat{\theta}$

and find ω_0 and β.

$F_\theta = mr\ddot{\theta} + 2\dot{r}\dot{\theta} = ml\ddot{\theta}$

$-mg\theta - bl\dot{\theta} = ml\ddot{\theta}$

$-g/l\,\theta - b/m\,\dot{\theta} = \ddot{\theta} \Rightarrow 0 = \ddot{\theta} + b/m\,\dot{\theta} + g/l\,\ddot{\theta} \Rightarrow \boxed{\beta = \frac{1}{2}\frac{b}{m},\ \omega_0 = \sqrt{g/l}}$

(b) Solve the equation of motion to find $\theta(t)$, assuming $\beta \neq \omega$. Use the same initial conditions as the previous problem: at $t = 0$, the pendulum is released from rest at a position θ_0 away from the vertical axis.

See Next Page

(c) What are the conditions on b to ensure that this pendulum oscillates? Write your answer in terms of m, g, and l.

We require $\sqrt{\beta^2 - \omega_0^2}$ to be imaginary

$\Rightarrow \beta < \omega_0 \Rightarrow \frac{1}{2}\frac{b}{m} < \sqrt{g/l} \Rightarrow \boxed{b < 2m\sqrt{g/l}}$

(d) What are the conditions on b to ensure that would mean this pendulum is not oscillating? Write your answer in terms of m, g, and l.

$\beta \geq \omega_0 \Rightarrow \boxed{b \geq 2m\sqrt{g/l}}$

(e) If $\boxed{\beta = \omega_0}$, find the solution for $\theta(t)$ using the same initial conditions.

In this case, $\theta(t) = e^{-\beta t} \neq 0$ only gives one solution

The other is $\theta(t) = te^{-\beta t}$

Check: $\dot{\theta} = e^{-\beta t} - \beta te^{-\beta t}$

$\ddot{\theta} = -\beta e^{-\beta t} - \beta e^{-\beta t} + \beta^2 te^{-\beta t}$

$\dddot{\theta} = -2\beta e^{-\beta t} + \beta^2 te^{-\beta t}$

$\omega_0^2 = \beta^2$

$\Rightarrow (-2\beta \cancel{e^{-\beta t}} + \beta^2 t \cancel{e^{-\beta t}}) + 2\beta(\cancel{e^{-\beta t}} - \beta t \cancel{e^{-\beta t}}) + \omega_0^2 t \cancel{e^{-\beta t}} = 0$

$-2\beta + \beta^2 t + 2\beta - 2\beta^2 t + \beta^2 t = 0$ ✓

cancel all cancel

$\theta(t) = A_1 e^{-\beta t} + A_2 te^{-\beta t}$

at $t=0$, $\theta = \theta_0$, $\dot{\theta} = 0$

$\Rightarrow \theta_0 = A_1$

$\dot{\theta} = -\beta A_1 e^{-\beta t} + A_2 e^{-\beta t} - \beta A_2 te^{-\beta t}$

$0 = -\beta A_1 + A_2 - 0 \Rightarrow A_2 = \beta A_1 = \beta\theta_0$

$$\boxed{\theta(t) = \theta_0 e^{-\beta t} + \beta\theta_0 te^{-\beta t}}$$

③ b) Check $\theta(t) = e^{rt}$

$$\Rightarrow 0 = r^2 e^{rt} + 2\beta r e^{rt} + \omega_0^2 e^{rt}$$

$$0 = r^2 + 2\beta + \omega_0^2 \Rightarrow r = -\beta \pm \sqrt{\beta^2 - \omega_0^2}$$

$$\Rightarrow \theta(t) = C_1 e^{(-\beta + \sqrt{\beta^2-\omega_0^2})t} + C_2 e^{(-\beta - \sqrt{\beta^2-\omega_0^2})t}$$

at $t=0$, $\theta(0) = \theta_0$:

$$\theta_0 = C_1 e^0 + C_2 e^0 = \boxed{C_1 + C_2 = \theta_0}$$

at $t=0$ $\dot{\theta}(0) = 0$:

$$\dot{\theta}(t) = C_1(-\beta + \sqrt{\beta^2-\omega_0^2})e^{(-\beta+\sqrt{\beta^2-\omega_0^2})t} + C_2(-\beta - \sqrt{\beta^2-\omega_0^2})e^{(-\beta-\sqrt{\beta^2-\omega_0^2})t}$$

$$0 = C_1(-\beta + \sqrt{\beta^2-\omega_0^2})e^0 + C_2(-\beta - \sqrt{\beta^2-\omega_0^2})e^0$$

$$0 = -\beta C_1 + C_1\sqrt{\beta^2-\omega_0^2} - \beta C_2 - C_2\sqrt{\beta^2-\omega_0^2}$$

$$0 = -\beta\underbrace{(C_1 + C_2)}_{\theta_0} + (C_1 - C_2)\sqrt{\beta^2-\omega_0^2}$$

$$\beta\theta_0 = (C_1 - C_2)\sqrt{\beta^2-\omega_0^2}$$

$$\left.\begin{aligned} C_1 - C_2 &= \frac{\beta\theta_0}{\sqrt{\beta^2-\omega_0^2}} \\ C_1 + C_2 &= \theta_0 \end{aligned}\right\} \text{add together}$$

$$\Rightarrow 2C_1 - C_2 + C_2 = \frac{\beta\theta_0}{\sqrt{\beta^2-\omega_0^2}} + \theta_0$$

$$C_1 = \frac{1}{2}\theta_0\left(1 + \frac{\beta}{\sqrt{\beta^2-\omega_0^2}}\right)$$

$$C_2 = \theta_0 - C_1 = \frac{1}{2}\theta_0(2) - \frac{1}{2}\theta_0\left(1 + \frac{\beta}{\sqrt{\beta^2-\omega_0^2}}\right)$$

$$= \frac{1}{2}\theta_0\left(2 - 1 - \frac{\beta}{\sqrt{\beta^2-\omega_0^2}}\right)$$

$$C_2 = \frac{1}{2}\theta_0\left(1 - \frac{\beta}{\sqrt{\beta^2-\omega_0^2}}\right)$$

$$\Rightarrow \theta(t) = \frac{1}{2}\theta_0\left[1 + \frac{\beta}{\sqrt{\beta^2-\omega_0^2}}\right]e^{(-\beta+\sqrt{\beta^2-\omega_0^2})t} + \frac{1}{2}\theta_0\left[1 - \frac{\beta}{\sqrt{\beta^2-\omega_0^2}}\right]e^{(-\beta-\sqrt{\beta^2-\omega_0^2})t}$$

Grade:

4. The Driven Damped Underwater Clock

Again consider that the water exerts a drag force on the pendulum shown in Fig. 2. This time, the pendulum is also subject to a driving force $F(t)$ in the $\hat{\theta}$ direction.

(a) Write down the equation of motion for the clock given both the drag and driving forces.

$F_\theta = -mg\theta - b\ell\dot{\theta} + F(t)$, $\quad F_\theta = m\ell\ddot{\theta}$

$\Rightarrow -mg\theta - b\ell\dot{\theta} + F(t) = m\ell\ddot{\theta}$

$\Rightarrow \ddot{\theta} + g/\ell\,\theta + b/m\,\dot{\theta} = F(t)$

$\ddot{\theta} + 2\beta\dot{\theta} + \omega_0^2\theta = f(t)$, $\quad f(t) = \frac{F(t)}{m\ell}$, $\quad \beta = \frac{b}{2m}$, $\quad \omega_0^2 = g/\ell$

(b) Let $F(t) = F_0\cos(\omega t)$ with the same initial conditions as the previous problem: at $t = 0$, the pendulum is released from rest at a position θ_0 away from the vertical axis.

i. Find $\theta(t)$.

See next page

ii. Find the value of ω that gives a resonance for this driven oscillator, assuming ω_0 is fixed.

$A^2 = \frac{f_0^2}{(\omega_0^2-\omega^2)^2 + 4\beta^2\omega^2}$, minimize denominator: $\frac{d}{d\omega}\left[(\omega_0^2-\omega^2)^2 + 4\beta^2\omega^2\right] = 2(\omega_0^2-\omega^2)(-2\omega) + 8\beta^2\omega$

$0 = -4\omega(\omega_0^2-\omega^2) + 8\beta^2\omega$

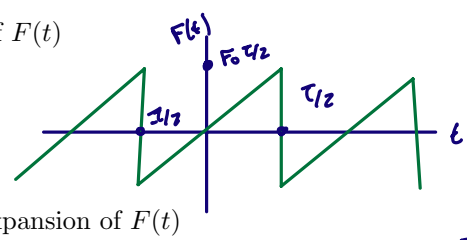
$0 = -4\omega_0^2 + 4\omega^2 + 8\beta^2 \Rightarrow \boxed{\omega^2 = \omega_0^2 - 2\beta^2}$

(c) Now consider instead that the driving force is periodic with the following shape:

$$F(t) = F_0 t, \qquad t \in (-\tau/2, \tau/2) \tag{1}$$

$$F(t+\tau) = F(t) \tag{2}$$

i. Sketch the shape of $F(t)$



ii. Find the Fourier expansion of $F(t)$

$F(t) = \text{odd} \Rightarrow a_0 = a_n = 0 \ \forall n$

$b_n = \frac{2}{\tau}\int_{-\tau/2}^{\tau/2} F(t)\sin n\omega t\,dt = \frac{2}{\tau}\int_{-\tau/2}^{\tau/2} F_0 t \sin n\omega t\,dt$

$= \frac{2}{\tau}F_0\left[-\frac{t}{n\omega}\cos n\omega t + \frac{1}{n^2\omega^2}\sin n\omega t\right]_{-\tau/2}^{\tau/2}$

$= \frac{2}{\tau}F_0\left[-\frac{\tau}{2n\omega}\cos\frac{n\omega\tau}{2} + \frac{1}{n^2\omega^2}\sin\frac{n\omega\tau}{2} - \frac{\tau}{2n\omega}\cos\frac{n\omega\tau}{2} - \frac{1}{n^2\omega^2}\sin\left(-\frac{n\omega\tau}{2}\right)\right]$

$b_n = \frac{2F_0}{\tau}\left[-\frac{\tau}{n\omega}\cos\frac{n\omega\tau}{2} + \frac{2}{n^2\omega^2}\sin\left(\frac{n\omega\tau}{2}\right)\right]$

$(\omega = 2\pi/\tau)$

$b_n = \frac{2F_0}{\tau}\left[-\frac{1}{2\pi n}\cos n\pi + \frac{2}{n^2\omega^2}\cancel{\sin n\pi}^{\,0}\right]$

$b_n = -\frac{F_0}{2\tau\pi n}(-1)^n$

$$\boxed{F(t) = \sum_{n=1}^{\infty}\frac{F_0}{2\tau\pi n}(-1)^n\sin n\omega t}$$

iii. Find the general solution for $\theta(t)$ given the driving force shown in Eq. (2).

$F(t) = \sum_n F_n \sin n\omega t \equiv \sum_n f_n(t)$

focus first on solving: $\ddot{x}_n + 2\beta x_n + \omega_0^2 x_n = f_n(t)/m$

Then the full $x(t) = \sum_n x_n(t)$ [By linearity of the EOM differential operator]

$F_n = \frac{F_0}{2\tau\pi n}(-1)^n$

from part (ii)

See next page

④ (b) $\ddot{\theta} + 2\beta\dot{\theta} + \omega_0^2\theta = f_0\cos\omega t$ where $f_0 = \frac{F_0}{m\ell}$

Also $\ddot{\phi} + 2\beta\dot{\phi} + \omega_0^2\phi = f_0\sin\omega t$ is a solution

Promote to complex variables: $\eta = \theta + i\phi$

$$\ddot{\eta} + 2\beta\dot{\eta} + \omega_0^2\eta = f_0 e^{i\omega t}$$

$$\eta_p(t) = Ce^{i\omega t}$$

$$-\omega^2 Ce^{i\omega t} + 2\beta i\omega Ce^{i\omega t} + \omega_0^2 Ce^{i\omega t} = f_0 e^{i\omega t}$$

$$C[-\omega^2 + 2i\omega\beta + \omega_0^2] = f_0$$

$$C = \frac{f_0}{\omega_0^2 - \omega^2 + 2i\beta\omega}$$

let $\underset{\text{complex}}{C} = \overset{\text{real}}{A}e^{-i\overset{\text{real}}{\delta}}$

$$\eta_p(t) = Ae^{i\omega t - i\delta}$$

$$\theta_p(t) = \mathrm{Re}(\eta_p(t)) = \mathrm{Re}\left[Ae^{i\omega t - i\delta}\right] = A\cos(\omega t - \delta)$$

where A, δ are on formula sheet

Remember $\theta_h(t) = C_1 e^{(-\beta + \sqrt{\beta^2 - \omega_0^2})t} + C_2 e^{(-\beta - \sqrt{\beta^2 - \omega_0^2})t}$

$$\theta_g(t) = A\cos(\omega t - \delta) + C_1 e^{(-\beta + \sqrt{\beta^2 - \omega_0^2})t} + C_2 e^{(-\beta - \sqrt{\beta^2 - \omega_0^2})t}$$

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④ (c) (iii)

$$F(t) = \sum_n F_n \sin n\omega t \equiv \sum_n f_n(t)$$

$$\ddot{x}_n + 2\beta\dot{x}_n + \omega_0^2 x_n = f_n(t) = \frac{F_n}{m\ell}\sin n\omega t \equiv f_n \sin n\omega t$$

let $\frac{F_n}{m\ell} \equiv f_n$ $\quad F_n = \frac{F_0}{2\pi\pi n}(-1)^n$ from part (ii)

$$\ddot{x}_n + 2\beta\dot{x}_n + \omega_0^2 x_n = f_n \sin n\omega t$$

let $z_n = y_n + ix_n$

$$\ddot{z}_n + 2\beta\dot{z}_n + \omega_0^2 z_n = f_n(\cos n\omega t + i\sin n\omega t) = f_n e^{in\omega t}$$

trial solution: $z_n = C_n e^{in\omega t}$

$$i^2 n^2\omega^2 C_n e^{in\omega t} + 2\beta in\omega C_n e^{in\omega t} + \omega_0^2 Ce^{in\omega t} = f_n e^{in\omega t}$$

$$C_n[-n^2\omega^2 + 2i\beta n\omega + \omega_0^2] = f_n$$

$$C_n = \frac{f_n}{\omega_0^2 - n^2\omega^2 + 2i\beta n\omega}$$

let $C_n = A_n e^{-i\delta_n}$, Find A_n, δ_n

$$C_n C_n^* = A e^{-i\delta} A e^{+i\delta} = A^2$$

$$C_n C_n^* = \frac{f_n}{\omega_0^2 - n^2\omega^2 + 2i\beta n\omega} \times \frac{f_n}{\omega_0^2 - n^2\omega^2 - 2i\beta n\omega}$$

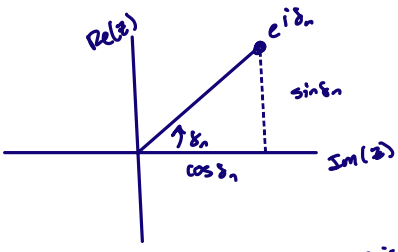
$$= \frac{f_n^2}{(\omega_0^2 - n^2\omega^2)^2 + 2i\beta n\omega(\omega_0^2 - n^2\omega^2) - 2i\beta n\omega(\omega_0^2 - n^2\omega^2) - 4i^2\beta^2 n^2\omega^2}$$

cancel

$$\boxed{A_n^2 = \frac{f_n^2}{(\omega_0^2 - n^2\omega^2)^2 + 4\beta^2 n^2\omega^2}}$$

Next: Find δ_n:

$$C_n = A_n e^{-i\delta_n}$$



$$\frac{1}{\omega_0^2 - n^2\omega^2 + 2i\beta n\omega} = A_n e^{-i\delta_n}$$

$$e^{i\delta} = A_n(\omega_0^2 - n^2\omega^2 + 2i\beta n\omega)$$

$$\mathrm{Re}[e^{i\delta_n}] = A(\omega_0^2 - n^2\omega^2)$$

$$\mathrm{Im}(e^{i\delta_n}) = 2A\beta n\omega$$

$$\tan\delta_n = \frac{\mathrm{Im}[e^{i\delta_n}]}{\mathrm{Re}[e^{i\delta_n}]}$$

$$\Rightarrow \boxed{\tan\delta_n = \frac{2\beta n\omega}{\omega_0^2 - n^2\omega^2}}$$

So $z_n(t) = C_n e^{in\omega t} = A_n e^{-i\delta_n} e^{in\omega t}$

Remember $z_n = y_n + i x_n$

$$x_n = \mathrm{Im}(z_n)$$

$$\Rightarrow x_n(t) = \mathrm{Im}\left[A_n e^{i(n\omega t - \delta_n)}\right]$$

$$x_n(t) = A_n \sin(n\omega t - \delta_n)$$

The general solution is therefore

$$x(t) = \sum_n x_n(t) = \sum_n A_n \sin(n\omega t - \delta_n)$$

$$x(t) = \sum_n \frac{f_n}{[(\omega_0^2 - n^2\omega^2)^2 + 4\beta^2 n^2\omega^2]^{1/2}} \sin\left[n\omega t - \tan^{-1}\left(\frac{2\beta n\omega}{\omega_0^2 - n^2\omega^2}\right)\right]$$

and $f_n = \frac{F_n}{\ell m} = \frac{F_0}{2\tau\pi n}(-1)^n \frac{1}{m\ell}$

$$x(t) = \sum_{n=1}^{\infty} \frac{F_0 (-1)^n}{2\tau\pi n m\ell} \frac{1}{[(\omega_0^2 - n^2\omega^2)^2 + 4\beta^2 n^2\omega^2]^{1/2}} \sin\left[n\omega t - \tan^{-1}\left(\frac{2\beta n\omega}{\omega_0^2 - n^2\omega^2}\right)\right]$$

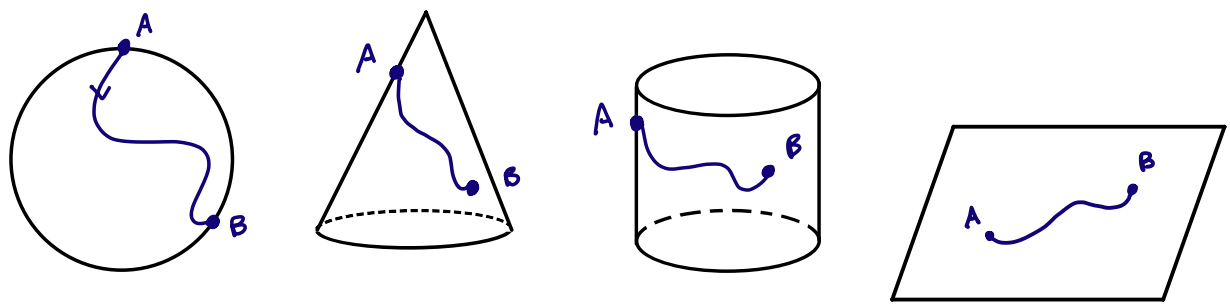
$\ldots = \theta(t)$ (or for switch notation)

Physics 3221 Spring 2026: Practice Midterm 2

Grade:

Note: this won't be on Midterm 2

5. ~~Geodesics~~ ~~Calculate the geodesic (the shortest distance between two points) of a few surfaces: cone, sphere, cylinder, plane~~Note: this won't be on Midterm 2



(a) A sphere with radius R

(b) A circular cone with a surface defined by $\rho = \lambda z$

(c) A cylinder of radius R

(d) A flat plane

Figure 3: Several surfaces

(a) Find the shortest distance between point A at position $(R, 0, 0)$ and B at position (R, θ_B, ϕ_B) following a path on the surface of a sphere shown in Fig. 3a.

(b) Find the shortest distance between point A at position (ρ_A, π, z_A) and B at position (ρ_B, ϕ_B, z_B) following a path on the surface of a cone shown in Fig. 3b. Once you get to trying to determine the two unknown coefficients by plugging in ϕ_b, z_b, z_A, you can stop, as the expressions get complicated.

(c) Find the shortest distance between point A at position (R, π, z_A) and B at position (R, ϕ_B, z_B) following a path on the surface of the cylinder shown in Fig. 3c.

(d) Find the shortest distance between point A at position $(0, 0, 0)$ and B at position (x_B, y_B, z_B) following a path on the surface of a cone shown in Fig. 3d.

Grade:

6. Things to Practice:

- Practice Line Integrals
- Practice Finding $U(\vec{r})$ from $\vec{F}(\vec{r})$ and vice versa
- Practice Checking Whether $\vec{F}$ is Conservative using Curl
- Practice Using Spherical Coordinates
- Practice Solving SHO 2nd order differential equations
- Practice Finding Fourier Series for various functions
- ~~Practice using calculus of variations to minimize actions~~

Formulas (just for the Practice Midterm)

Newton's 2nd Law

$$\vec{F} = m\ddot{\vec{r}}$$

Cartesian

$$F_x = m\ddot{x} \qquad F_y = m\ddot{y} \qquad F_z = m\ddot{z}$$

Polar

$$F_r = m(\ddot{r} - r\dot{\phi}^2) \qquad F_\phi = m(r\ddot{\phi} + 2\dot{r}\dot{\phi})$$

Cylindrical

$$F_\rho = m(\ddot{\rho} - \rho\dot{\phi}^2) \quad F_\phi = m(\rho\ddot{\phi} + 2\dot{\rho}\dot{\phi}) \quad F_z = m\ddot{z}$$

Coordinate Systems

Cartesian: $\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$
$d\vec{r} = dx\hat{x} + dy\hat{y} + dz\hat{z}$

Cylindrical : $\vec{r} = \rho\hat{\rho} + z\hat{z}$
$d\vec{r} = d\rho\hat{\rho} + \rho d\phi\hat{\phi} + dz\hat{z}$

Spherical: $\vec{r} = r\hat{r}$
$d\vec{r} = dr\hat{r} + rd\theta\,\hat{\theta} + r^2\sin\theta\, d\phi\,\hat{\phi}$

Euler's Formula

$$e^{i\theta} = \cos\theta + i\sin\theta$$

Fourier Series

$$f(t) = \sum_{n=0}^{\infty}\left[a_n\cos(n\omega t) + b_n\sin(n\omega t)\right],$$

$$\omega = 2\pi/\tau.$$

$$a_0 = \frac{1}{\tau}\int_{-\tau/2}^{\tau/2} f(t)\,,$$

$$a_n = \frac{2}{\tau}\int_{-\tau/2}^{\tau/2} f(t)\cos(n\omega t)dt\,, \qquad n > 0$$

$$b_n = \frac{2}{\tau}\int_{-\tau/2}^{\tau/2} f(t)\sin(n\omega t)dt\,, \qquad n > 0$$

Hook's Law & Simple Harmonic Motion

$$F = -kx \Leftrightarrow U = \frac{1}{2}kx^2$$

$$x(t) = A\cos(\omega t - \delta)$$

Driven Oscillator

$$A^2 = \frac{f_0^2}{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2)}$$

$$\delta = \tan^{-1}\left(\frac{2\beta\omega}{\omega_0^2 - \omega^2}\right)$$

Lorentz Force Law

$$\vec{F} = q(\vec{E} + \vec{v}\times\vec{B})$$

Work

$$W = \int_{\vec{r}_0}^{\vec{r}} \vec{F}(\vec{r}\,')\cdot d\vec{r}\,'$$

Conservative forces

$$U(\vec{r}) = -\int_{\vec{r}_0}^{\vec{r}} \vec{F}(\vec{r}\,')\cdot d\vec{r}\,'$$

$$F = -\vec{\nabla}U$$

$$\vec{\nabla}\times\vec{F} = 0$$

Useful Integrals

$$\int x\sin(ax)dx = -\frac{x}{a}\cos(ax) + \frac{1}{a^2}\sin(ax)$$

$$\int x\cos(ax)dx = \frac{x}{a}\sin(ax) + \frac{1}{a^2}\cos(ax)$$

Gradient and Curl

$$\vec{\nabla}f = \hat{x}\frac{\partial f}{\partial x} + \hat{y}\frac{\partial f}{\partial y} + \hat{z}\frac{\partial f}{\partial z}$$

$$\vec{\nabla}f = \hat{r}\frac{\partial f}{\partial r} + \hat{\theta}\frac{1}{r}\frac{\partial f}{\partial\theta} + \hat{\phi}\frac{1}{r\sin\theta}\frac{\partial f}{\partial\phi}$$

$$\vec{\nabla}f = \hat{\rho}\frac{\partial f}{\partial r\rho} + \hat{\theta}\frac{1}{\rho}\frac{\partial f}{\partial\theta} + \hat{z}\frac{\partial f}{\partial z}$$

$$\vec{\nabla}\times\vec{A} = \hat{x}\left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z}\right) + \hat{y}\left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x}\right) + \hat{z}\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)$$